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Case 3-2022: A 14-Year-Old Boy with Fever, Joint Pain, and Abdominal Cramping

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PRESENTATION OF CASE

Dr. Micaela Atkins (Pediatrics): A 14-year-old boy was admitted to this hospital because of fatigue, fever, joint pain, abdominal cramping, and diarrhea.

The patient had been well until 2 weeks before this admission, when fatigue and fever developed on his final day of summer camp. He was taken to the primary care clinic at another hospital and was told that he had a viral infection.

Nine days before this admission, new mild sore throat developed, and the patient returned to the primary care clinic. A throat culture for group A beta-hemolytic streptococcus was reportedly negative. The patient was told that he had possible sinusitis, and treatment with amoxicillin–clavulanate was started. During the next 3 days, the sore throat abated, but fatigue and fever persisted.

Six days before this admission, new pain in the right shoulder and left knee developed, and the patient again returned to the primary care clinic. The white-cell count and erythrocyte sedimentation rate were reportedly elevated; a blood test for Lyme disease was performed.

On the day of admission, the joint pain progressed to involve the shoulders and knees on both sides, and the temperature increased to 39.4°C. The patient was given ibuprofen and brought to the emergency department at this hospital. On evaluation, the patient described aching pain in the shoulders and knees, which was worst in his right shoulder and left knee. He rated the pain at 7 on a scale of 0 to 10, with 10 indicating the most severe pain. He had not noticed redness, swelling, or stiffness of the joints. A review of systems was notable for chills, intermittent night sweats, headache, myalgias, and lightheadedness when he stood up from a sitting position. He had no weight loss, rash, vision changes, or respiratory symptoms. He had mild abdominal cramping, decreased appetite, and intermittent nausea. During the past week, there had been a few episodes of nonbloody emesis and watery diarrhea. There had been no hematemesis, hematochezia, or melena.

The patient had autism spectrum disorder, with an early delay in speech development that had resolved after speech therapy. He had met milestones for gross

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Table 1. Laboratory Data.*

Variable	Reference Range†	On Admission, This Hospital
Hemoglobin (g/dl)	13.0–16.0	12.6
Hematocrit (%)	37.0–49.0	38.9
Platelet count (per μ l)	150,000–450,000	419,000
White-cell count (per μ l)	4500–13,000	12,950
Differential count (per μ l)		
Neutrophils	1800–8100	8530
Lymphocytes	1200–5200	2330
Monocytes	300–1400	1640
Eosinophils	0–1000	350
Basophils	0–400	50
Erythrocyte sedimentation rate (mm/hr)	0–13	48
C-reactive protein (mg/liter)	0.0–8.0	207.6
Alanine aminotransferase (U/liter)	10–55	9
Aspartate aminotransferase (U/liter)	10–40	9
Alkaline phosphatase (U/liter)	55–149	81
Lipase (U/liter)	13–60	82
Albumin (g/dl)	3.3–5.0	3.1
Sodium (mmol/liter)	135–145	138
Potassium (mmol/liter)	3.4–5.0	3.8
Chloride (mmol/liter)	98–108	102
Carbon dioxide (mmol/liter)	23–32	26
Urea nitrogen (mg/dl)	8–25	8
Creatinine (mg/dl)	0.60–1.50	0.72

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

and fine motor skills and growth. He had reportedly received all routine childhood vaccinations. Other history included asthma, seasonal rhinosinusitis, and allergies to peanuts and tree nuts; there were no known drug allergies. He had undergone a tonsillectomy in early childhood. Medications included amoxicillin–clavulanate, inhaled fluticasone, and ibuprofen and inhaled albuterol as needed.

At the time of admission, the patient was on his summer break before entering high school. Earlier in the summer, he had gone on vacation to a coastal area of New England. He had also attended camp in a rural area of New England,

where he camped and hiked in wooded areas and swam in fresh water. He had seen a tick on his clothing but did not recall any bites. Two weeks before this admission, the patient had returned to his home in a suburban area of New England, where he lived with his mother (who was a veterinarian), father, sister, and pet dog. His sister had recently had an acute gastrointestinal illness that resolved after several days. The patient was not sexually active and did not use alcohol, drugs, or tobacco. His mother had hypothyroidism, and his maternal uncle had rheumatoid arthritis; there was no other known family history of autoimmune diseases.

On examination, the temperature was 36.1°C, the blood pressure 128/58 mm Hg, the heart rate 107 beats per minute, the respiratory rate 18 breaths per minute, and the oxygen saturation 97% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 27.2. The patient appeared to be well developed and well nourished, but he looked tired. The conjunctivae were normal. The mucous membranes were dry, and there was an aphthous ulcer on the right buccal mucosa. The lungs were clear on auscultation. There was tenderness on palpation of the epigastric and periumbilical abdomen. There was no erythema, warmth, swelling, or crepitus of the large and small joints. There was tenderness on palpation, as well as pain with movement, of the right shoulder and knees; range of motion in these joints was not decreased. The patient had no rash or lymphadenopathy.

The white-cell count was 12,950 per microliter (reference range, 4500 to 13,000), the erythrocyte sedimentation rate 48 mm per hour (reference range, 0 to 13), the C-reactive protein level 207.6 mg per liter (reference range, 0.0 to 8.0), and the lipase level 82 U per liter (reference range, 13 to 60). Laboratory test results are shown in Table 1.

While the patient was in the emergency department, the temperature increased to 39.2°C. Intravenous fluids were administered, and empirical treatment with doxycycline was started. The patient was admitted to the hospital.

During the next 2 days, the fever resolved, but the joint pain, abdominal cramping, and diarrhea worsened in severity. On the third hospital day, fever recurred. The white-cell count was

15,190 per microliter, the C-reactive protein level 178.3 mg per liter, and the lipase level 145 U per liter. A fecal occult blood test was positive. A plain film of the right shoulder was normal.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Kerstin Zanger: This previously healthy 14-year-old boy presented with fatigue, fever, sore throat, joint pain, abdominal cramping, nausea, vomiting, and watery diarrhea. The laboratory test results were notable for elevated levels of inflammatory markers and mild leukocytosis with neutrophilic predominance. Among his mainly nonspecific symptoms, the two that stand out are fever and joint pain. As I develop a differential diagnosis, I will focus on infectious and noninfectious causes of these symptoms.

INFECTIOUS CAUSES

Joint Infection

Patients with septic arthritis have an acute presentation with high-grade fevers and, usually, a single warm and swollen joint, most commonly in the lower leg. Approximately 80% of cases involve the knee, hip, or ankle.¹ Up to 10% of cases involve more than one joint. When multiple joints are involved, infection with *neisseria* species or *Staphylococcus aureus* must be considered. Patients with septic arthritis usually have decreased range of motion and loss of function in the affected joint, as well as extreme pain with both active and passive movement of the joint, features that were not present in this patient.

Lyme Disease

Lyme disease is a tickborne infection caused by *Borrelia burgdorferi*. Symptoms of early localized Lyme disease, which develop within a few days after the tick bite, include fatigue, headaches, myalgias, arthralgias, and fevers.² Erythema migrans occurs in approximately 80% of patients, so the absence of a rash does not rule out the infection. Arthralgias occur in 33% of patients during early disease and may be the only manifestation of the infection. Joint pain is often transient and migratory; it usually lingers at one site for hours or days. Gastrointestinal symptoms are uncommon during early localized Lyme disease, but they can be part of the initial influenza-like illness associated with the infection.

This patient lived in an area where Lyme disease is endemic and had been camping in the woods, so this infection must be considered. However, he had been treated with amoxicillin-clavulanate for 12 days and with doxycycline for 3 days; both of these agents are highly effective against Lyme disease. Antibiotic resistance in *B. burgdorferi* has not been observed.³

Reactive and Viral Arthritis

Reactive arthritis occurs in response to an infection that developed days or weeks earlier. In children, the most common cause is an enteric bacterial infection due to salmonella, shigella, yersinia, or campylobacter species or *Escherichia coli*. Such an infection would cause bloody diarrhea and abdominal pain and would be associated with positive fecal occult blood testing. After an enteric infection, enthesitis is more common than arthritis.

Arthritis due to parvovirus B19 occurs in less than 10% of infected children, as compared with 60% of infected adults. Children with arthritis due to parvovirus B19 initially have mild upper respiratory symptoms, followed by fever, rash, and polyarthritis. Levels of inflammatory markers may be mildly elevated but are normal in more than 50% of cases. Fevers are usually low grade.

This patient's arthralgias occurred concurrently with gastrointestinal symptoms. In addition, he had high-grade fevers and elevated levels of inflammatory markers. Therefore, a diagnosis of reactive or viral arthritis is unlikely.

Whipple's Disease

Whipple's disease is an enteric infection due to *Tropheryma whippeli*. The disease is not associated with a clear inflammatory and immunologic response. However, it results in clinical symptoms that are consistent with this patient's presentation, including arthritis, intermittent diarrhea, and colicky abdominal pain. It has been described as a mimicker of inflammatory bowel disease (IBD).⁴ Whipple's disease usually affects White middle-aged men of European ancestry and is extremely uncommon in children. Yet cases of acute gastroenteritis caused by infection with *T. whippeli* have been described in very young children in Europe and Africa.⁵ Because Whipple's disease is not associated with fever and is a rare condition, it can be ruled out in this patient.

NONINFECTIOUS CAUSES*Systemic Juvenile Idiopathic Arthritis*

Systemic juvenile idiopathic arthritis is characterized by fever with a daily spike in temperature ($>38.5^{\circ}\text{C}$) and with normal or subnormal temperatures during the rest of the 24-hour period. Affected patients often appear ill when they are febrile and otherwise appear well. This condition is also characterized by arthritis, which can involve any joint and any number of joints but most often occurs in the lower leg.⁶ In 81% of patients, an evanescent salmon-pink rash develops, although it is sometimes difficult to see. The rash usually is triggered by heat or fever and fades when the body temperature is normal. It often involves the axillae and the area around the waist but can occur anywhere on the body. Patients with systemic juvenile idiopathic arthritis typically have leukocytosis with a white-cell count of 20,000 to 30,000 per microliter and neutrophilic predominance.

This patient's fever did not occur on a daily basis, and he did not have a rash. In addition, to establish the diagnosis of systemic juvenile idiopathic arthritis, fever must be present for at least 2 weeks and arthritis for at least 6 weeks. Therefore, the diagnosis can be ruled out in this patient.

Inflammatory Bowel Disease

Most children with a new diagnosis of IBD present with bloody diarrhea, abdominal pain, tenesmus, and growth failure. Arthritis and arthralgias are frequent extraintestinal manifestations; they occur more commonly in patients who have Crohn's disease with colonic involvement than in patients who have ulcerative colitis, and they are often associated with other extraintestinal features, such as aphthous stomatitis.⁷ Approximately 6% of pediatric patients have extraintestinal symptoms at the time of initial presentation. Arthritis due to IBD is self-limiting and nonerosive, and it is not associated with specific autoantibodies or HLA types. Peripheral arthritis tends to be acute, occurs early in the disease course, and involves few joints. There is often evidence of joint swelling and effusion, most commonly involving the knee. The shoulder is affected in only 4 to 9% of patients.⁸

Elevations in the white-cell count and levels of inflammatory markers reflect the activity of intestinal disease and not the severity of arthri-

tis. Approximately one third of pediatric patients with IBD have a normal white-cell count and normal levels of inflammatory markers at the time of diagnosis.⁹ Children with IBD often "fall off" the growth curve long before the diagnosis is made. An early sign of growth failure is a decrease in height velocity.

The most likely diagnosis in this patient is IBD. His initial vague gastrointestinal symptoms are atypical of this condition; however, arthralgias and oral aphthous ulcers are both common extraintestinal manifestations. He did not have frankly bloody stools, which could explain why the hemoglobin level was only mildly decreased. The decreased albumin level and mildly elevated lipase level are consistent with a diagnosis of IBD, and the elevated levels of inflammatory markers support the diagnosis.

Several features of this patient's presentation give me pause. Presentation with arthralgias as the dominant symptom and with only mild gastrointestinal symptoms is highly unusual, as is the fact that the most affected joint was his shoulder. Also, he did not have a history of weight loss or growth failure; the absence of these features suggests a recent and rapid onset of illness. Despite his somewhat unusual presentation, I think IBD is the most likely diagnosis in this case. To establish this diagnosis, I would perform colonoscopy and upper endoscopy.

DR. KERSTIN ZANGER'S DIAGNOSIS

Inflammatory bowel disease.

DIAGNOSTIC TESTING

Dr. Katherine Nimkin: The first diagnostic test was computed tomographic enterography (CTE), performed after the administration of oral and intravenous contrast material. Coronal images (Fig. 1) showed mural thickening and enhancement involving at least 20 cm of distal ileum, a finding consistent with ileitis. The comb sign — dilated, widely spaced, enhancing vasa recta — was present, a finding that reflects bowel inflammation.

Ileitis can be associated with a variety of conditions. Crohn's disease is usually the leading diagnosis. Patients with ulcerative colitis may have distal ileitis, referred to as backwash ileitis. Reactive ileitis can occur when there is an adja-

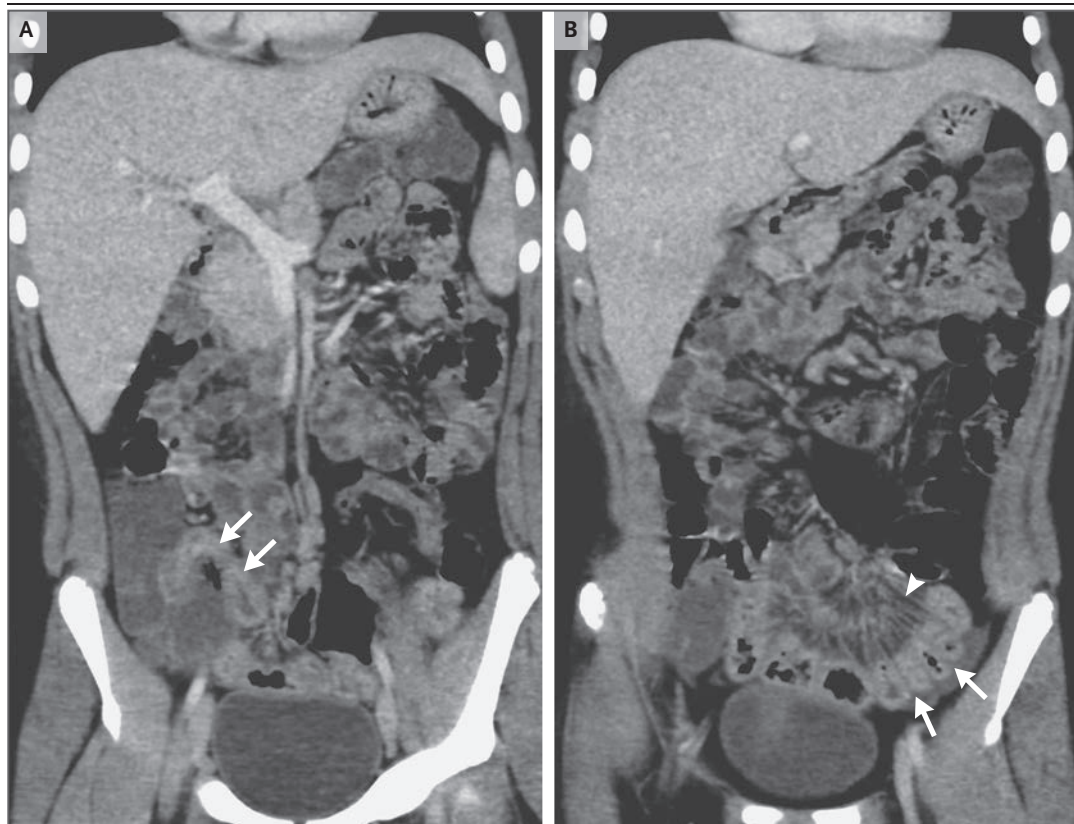


Figure 1. Images from CT Enterography.

In Panels A and B, coronal images show mural enhancement and mild thickening (arrows) involving approximately 20 cm of distal ileum, extending to the terminal ileum. Panel B also shows engorged vasa recta (arrowhead), known as the comb sign, a finding that reflects hypervascularity of the mesentery.

cent inflammatory process, such as appendicitis. Infectious ileitis can be caused by a variety of bacterial and viral pathogens; tuberculous ileitis may be seen in areas where the pathogen is endemic. Rarer causes include lymphoma, ischemia, vasculitis, and the use of nonsteroidal anti-inflammatory drugs.¹⁰

When Crohn's disease is suspected, cross-sectional imaging is indicated. Ultrasonography is performed at some centers, but the examination is operator dependent and time consuming. CTE and magnetic resonance enterography (MRE) can be used in the evaluation for Crohn's disease. Both techniques involve the ingestion of a large volume of oral contrast material, as well as the use of intravenous contrast material. It is appropriate to perform CTE as the first cross-sectional examination or when an abscess or perforation is suspected. CTE is fast and easily accessible, and young patients are less likely to

be sedated during CTE than during MRE. In comparison, MRE is performed without the use of ionizing radiation and provides a more detailed evaluation of disease activity and bowel motility. Magnetic resonance imaging of the pelvis is the study of choice in the evaluation for suspected perianal disease.

Features of active Crohn's disease on CTE and MRE include mural thickening and enhancement, with a striated pattern of mural enhancement, as well as engorged vasa recta. Both techniques can be used to assess for stricture formation, fistulae, and abscesses and to detect extraintestinal disease, such as sacroiliitis or sclerosing cholangitis.¹¹

Dr. Esther J. Israel: After the patient underwent CTE, endoscopy was performed. On endoscopic examination, the colon appeared normal, apart from the presence of a few scattered white lesions with red bases, which were suggestive of

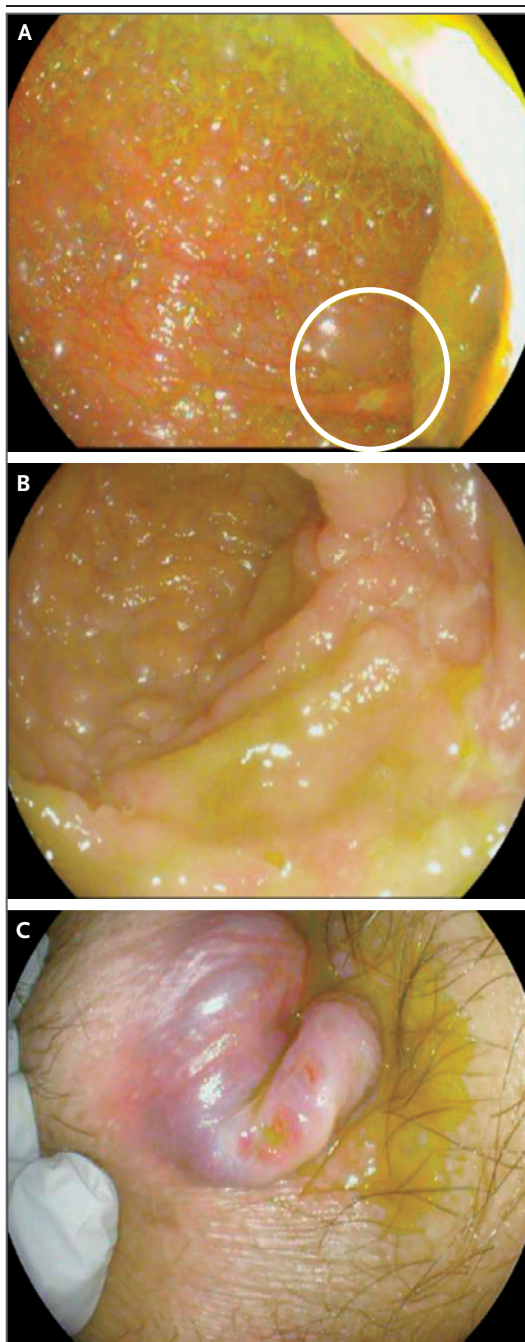


Figure 2. Images from Endoscopy.

In Panel A, the colon appears normal, apart from the presence of a few scattered white lesions with red bases (circle), which are suggestive of aphthous ulcers. In Panel B, the terminal ileum shows mild diffuse inflammation that is characterized by erosions, erythema, loss of vascularity, mucus, and aphthous ulceration. In Panel C, the perianal area shows a prominent hemorrhoid.

aphthous ulcers (Fig. 2A). The terminal ileum showed mild diffuse inflammation that was characterized by erosions, erythema, loss of vascularity, mucus, and aphthous ulceration (Fig. 2B). Biopsy specimens were obtained. On perianal examination, hemorrhoids were present (Fig. 2C).

PATHOLOGICAL DISCUSSION

Dr. Angela R. Shih: Histologic examination of a biopsy specimen from the ileum revealed ulcerated intestinal mucosa with fibrinopurulent exudate and blunted enteric villi (Fig. 3A). The underlying lamina propria showed an infiltrate of scattered neutrophils between residual glands, a finding consistent with severely active ileitis (Fig. 3B). The lamina propria also showed a focus with loose aggregates of histiocytes, a finding consistent with granulomatous inflammation (Fig. 3C). No definitive features of chronicity were identified.

Histologic examination of a biopsy specimen from the stomach revealed gastric antral mucosa with mild expansion of the lamina propria by a lymphoplasmacytic inflammatory infiltrate, a finding consistent with chronic inactive gastritis (Fig. 3D). No organisms were identified. Histologic examination of biopsy specimens from the esophagus, duodenum, and colon showed no abnormalities in the esophageal squamous mucosa, duodenal mucosa, and colonic mucosa, respectively.

Taken together, the severely active ileitis and granulomatous inflammation are consistent with early IBD, with a pattern of involvement and histologic features suggestive of Crohn's disease. Mild chronic inactive gastritis may also be seen in patients with this condition.

PATHOLOGICAL DIAGNOSIS

Inflammatory bowel disease (Crohn's disease).

DISCUSSION OF MANAGEMENT

Dr. Israel: In this 14-year-old boy with a new diagnosis of Crohn's disease, we considered three major goals of management: alleviation of symptoms, healing of mucosa, and improvement of his overall quality of life. Ideally, the chosen treatment regimen would address all three goals.

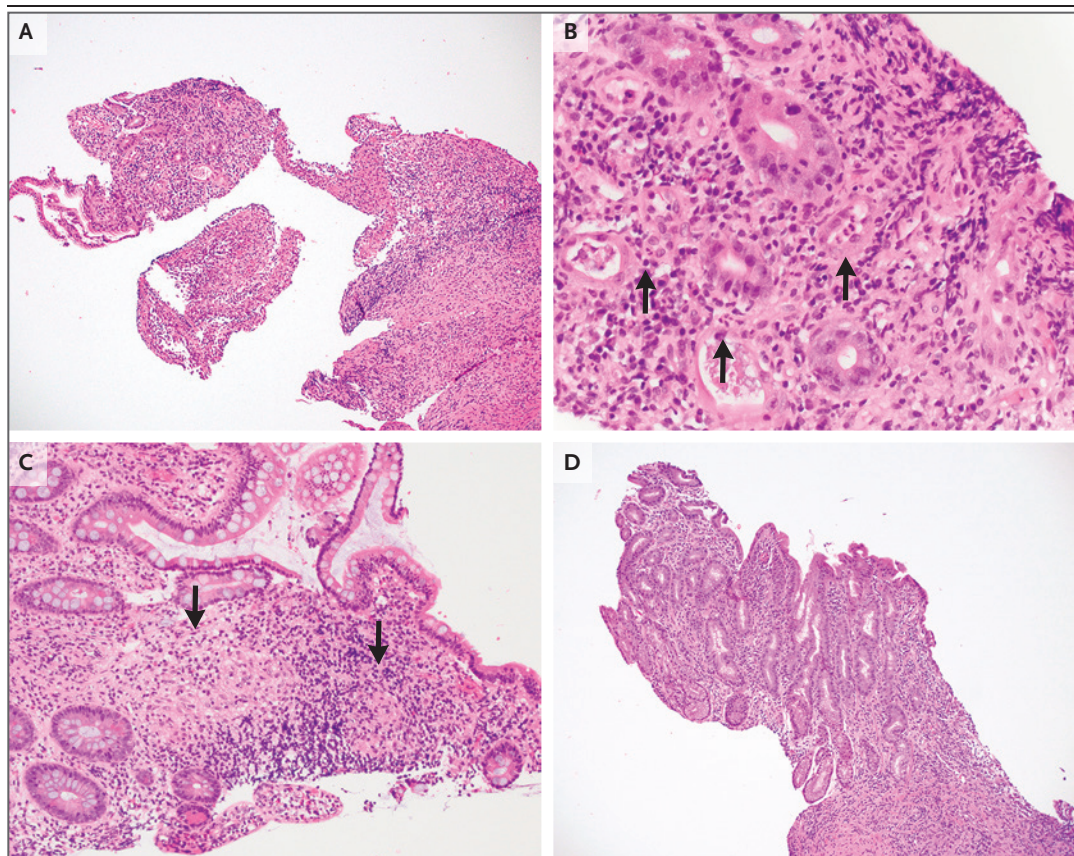


Figure 3. Biopsy Specimens from the Ileum and Stomach.

On hematoxylin and eosin staining, a biopsy specimen from the ileum shows ulcerated mucosa with fibrinopurulent exudate and blunted enteric villi (Panel A). At higher magnification, the lamina propria shows neutrophilic inflammation (Panel B, arrows), a finding consistent with severely active ileitis. The lamina propria also shows focal granulomatous inflammation (Panel C, arrows). A biopsy specimen from the stomach shows gastric antral mucosa with mild expansion of the lamina propria by a lymphoplasmacytic inflammatory infiltrate (Panel D), a finding consistent with mild chronic inactive gastritis. Biopsy specimens from the esophagus, duodenum, and colon showed no abnormalities (not shown). The overall findings are consistent with early inflammatory bowel disease, with histologic features suggestive of Crohn's disease.

Mucosal healing is the primary goal of treatment, because the correlation of clinical symptoms and laboratory values with healing can be poor. Clinical symptoms and abnormal laboratory values may abate before mucosal healing is complete, so they are not the most reliable markers of treatment response.¹²

NUTRITIONAL THERAPY

In patients with Crohn's disease, nutritional therapy is used to attain adequate growth and metabolic balance, control symptoms, and treat inflammation. The consensus guidelines of the European Crohn's and Colitis Organisation and

the European Society for Paediatric Gastroenterology Hepatology and Nutrition¹³ indicate that, in children with active luminal Crohn's disease, the recommended first-line treatment for inducing remission is dietary therapy with exclusive enteral nutrition, which provides at least 90% of calories as liquid. This therapy is rarely used in the United States, although it is the standard treatment for patients with newly diagnosed luminal Crohn's disease in Canada and Europe. Exclusive enteral nutrition and glucocorticoid therapy are associated with similar rates of remission, and exclusive enteral nutrition appears to be superior to glucocorticoid therapy in terms

of bringing about mucosal healing.¹⁴ In addition, exclusive enteral nutrition is equivalent to anti-tumor necrosis factor α (TNF- α) therapy in terms of controlling symptoms.¹⁵ For maintenance, a liquid diet is untenable, but two diets — the specific carbohydrate diet and the Crohn's disease exclusion diet — have been shown to be effective in keeping patients in remission.¹⁶⁻¹⁹ There is some evidence that these diets can even be used during the induction period. Both diets require intensive nutritional counseling, because the details are highly specific.

MEDICATIONS

When exclusive enteral nutrition is not an option, guidelines support the use of glucocorticoids for inducing remission in children with active luminal Crohn's disease. Although glucocorticoids are often very potent for induction therapy, they cannot be used for maintenance therapy because of side effects; they are used only as a bridge to maintenance therapy. After glucocorticoid treatment, maintenance therapy might include immunomodulators such as mercaptopurine, azathioprine, or methotrexate or long-term nutritional therapy such as the specific carbohydrate diet. There are concerns regarding the development of an aggressive hepatosplenic T-cell lymphoma in adolescent boys and young men who received either mercaptopurine or azathioprine, so we rarely use these two medications at our institution. Acetylsalicylic acid compounds are sometimes used to treat mild luminal disease, although there is scant evidence regarding their efficacy in maintaining remission and achieving mucosal healing.

In patients with a high risk of a complicated disease course, anti-TNF- α therapy is recommended for both induction and maintenance therapy. Factors that have been associated with an increased risk of progression of Crohn's disease are a young age at presentation, the presence of deep colonic ulcerations on endoscopic examination, extensive small-bowel disease, marked growth retardation, the presence of strictures and penetrating disease at onset, and perianal disease.²⁰ Anti-TNF- α therapy is highly effective in maintaining remission and achieving mucosal healing. If the patient does not have a response to anti-TNF- α therapy or if the anti-TNF- α agents cause a hypersensitivity reaction or lose their efficacy, treatment with ustekinumab

(antibodies against interleukin-12 and interleukin-23)²¹ or vedolizumab (antibodies against integrins)²² can be considered.

ADDITIONAL CONSIDERATIONS

Among patients with IBD, fatigue is one of the most frequently reported concerns. The following algorithm can be used for the management of fatigue: assess for adverse drug effects; treat iron and vitamin B₁₂ deficiencies and correct anemia; optimize therapy for active inflammation; treat mood and sleep disorders with the use of pharmacologic, behavioral, and psychological interventions; and encourage physical activity.²³

In the management of any chronic illness, psychological and behavioral health need to be considered. Among children and young adults with IBD, the risk of a psychiatric disorder is four times as high as the risk in the general population.^{24,25} The team that provides care for a child with IBD should include a mental health professional.

FOLLOW-UP

Dr. Israel: Although this patient's parents wanted him to start nutritional therapy during the induction phase, the patient declined. He therefore started treatment with intravenous methylprednisolone, which was transitioned to oral prednisone. The plan was to transition to an acetylsalicylic acid agent during the maintenance phase, but his symptoms returned when the dose of prednisone was tapered, and he started treatment with mesalamine. The fecal calprotectin level and levels of other inflammatory markers remained elevated.

Approximately 4 months after diagnosis, the patient agreed to try nutritional therapy. He received exclusive enteral nutrition for 4 days and then followed the specific carbohydrate diet for 2 weeks, but he did not wish to continue. Treatment with the anti-TNF- α agent adalimumab was started. The erythrocyte sedimentation rate, C-reactive protein level, and fecal calprotectin level decreased, and the patient appeared to have a good response. However, within 3 months, the trough level of the medication was low and antibody levels were very high.

Treatment was therefore switched to the anti-TNF- α agent infliximab, and methotrexate was

added to decrease immunogenicity. The patient received that regimen for the next 1.5 years. During treatment, he reported that he still had fatigue and hematochezia and was not feeling well. However, the erythrocyte sedimentation rate, C-reactive protein level, and fecal calprotectin level remained normal.

One year after the initiation of infliximab therapy, colonoscopic examination showed mucosal healing in the colon and MRE showed decreased disease activity in the ileum. However, 1.5 years after the initiation of infliximab therapy, the patient had a reaction during an infusion. Although we considered trying another anti-TNF- α agent, the patient wanted to change the class of medication because he was not feeling better. He started treatment with ustekinumab, and within a couple of months, symptoms wors-

ened, the fecal calprotectin level increased, and a perianal abscess developed. Treatment was then changed to the anti-TNF- α agent certolizumab, but high antibody levels quickly developed. He is now being desensitized to infliximab, because treatment with that agent seemed to have the best results and anti-TNF- α agents appear to be the most effective in treating perianal disease, which he now has. Throughout this time, he has been regularly seeing a mental health professional.

FINAL DIAGNOSIS

Inflammatory bowel disease (Crohn's disease).

This case was presented at Pediatric Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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